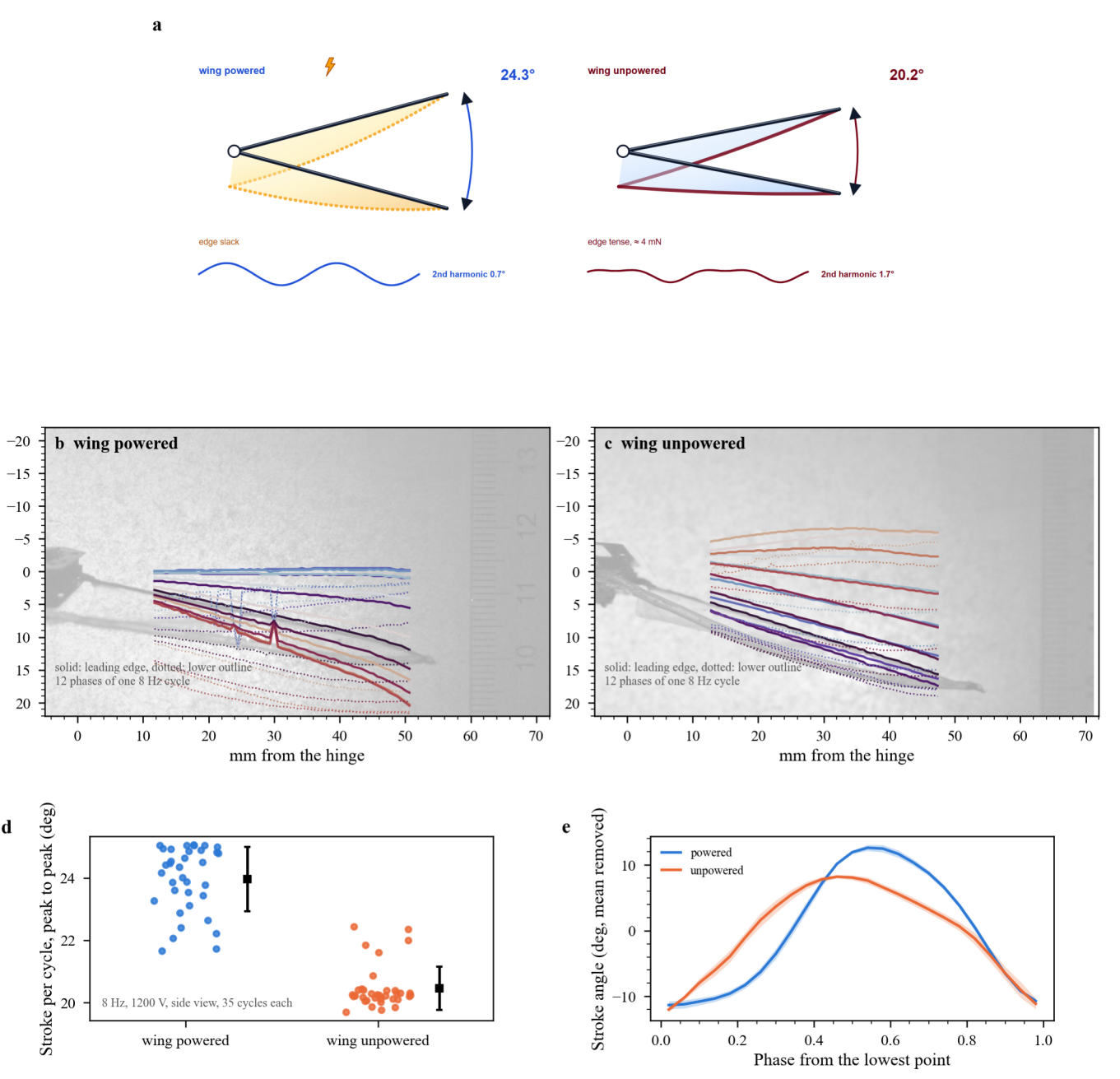

a
wing powered
24.3°
wing unpowered
20.2°
edge slack
2nd harmonic 0.7°
edge tense, ≈ 4 mN
2nd harmonic 1.7°
b wing powered
solid: leading edge, dotted: lower outline
12 phases of one 8 Hz cycle
mm from the hinge
c wing unpowered
solid: leading edge, dotted: lower outline
12 phases of one 8 Hz cycle
mm from the hinge
d
Stroke per cycle, peak to peak (deg)
8 Hz, 1200 V, side view, 35 cycles each
wing powered
wing unpowered
e
Stroke angle (deg, mean removed)
powered
unpowered
Phase from the lowest point